Work Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links work (W), force (F), and distance (d). Cover the unknown to read the formula: $W = F \times d$, $F = W \div d$, and $d = W \div F$.

Round 1 — Retrieve It

1. What three quantities does the work triangle connect?
2. Write the formula for work.
3. Write the formula for force.
4. Which variable do you cover to find distance, and what formula results?

Round 2 — Sketch It

Draw the work triangle. Put work on top; put force and distance on the bottom. Circle the two quantities you would multiply to find work.

Work — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A force of 5 N pushes a box 4 m across the floor. How much work is done?
2. A crane does 60 J of work lifting a load 3 m. What force does it apply?
3. A worker does 90 J of work pushing a cart with a force of 15 N. How far does the cart move?

Round 4 — Reason from Evidence

Two students each push with the same force of 10 N. Student A pushes a cart 2 m; Student B pushes an identical cart 5 m. Use the triangle to find the work each student does, then say which student does more work and how you know.

Work — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A Articulate

Explain how the work triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C Connect

Connect work to the energy transferred to an object. Explain why no work is done if there is no movement in the direction of the force, even when you push hard.

E Extend

Create your own problem that asks for distance. Give the numbers and show the work using $d = W \div F$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Work

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Work, force, and distance.
2. $W = F \times d$.
3. $F = W \div d$.
4. Cover d; $d = W \div F$.

Sketch It — look for

Work on top, with force and distance on the bottom. Work is found by multiplying force × distance (the two bottom quantities).

Apply It

1. $W = F \times d = 5 \times 4 =$ **20 J**.
2. $F = W \div d = 60 \div 3 =$ **20 N**.
3. $d = W \div F = 90 \div 15 =$ **6 m**.

Reason from Evidence — look for

Student A: $W = 10 \times 2 = 20$ J. Student B: $W = 10 \times 5 = 50$ J. Student B does more work because, with the same force, moving the object a greater distance transfers more energy.

ACE — Extend look-for

Many answers possible. Check that the problem asks for distance and the work correctly uses $d = W \div F$.

> **Suggested use.** Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.